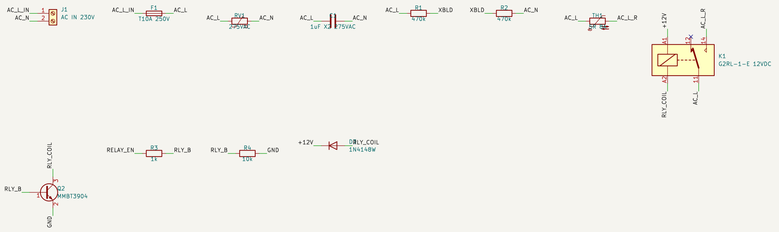
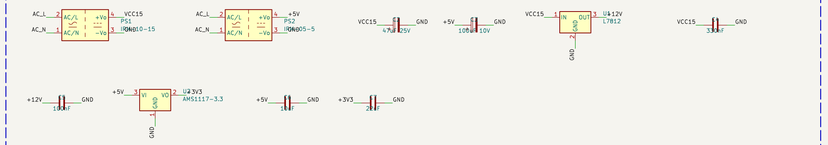
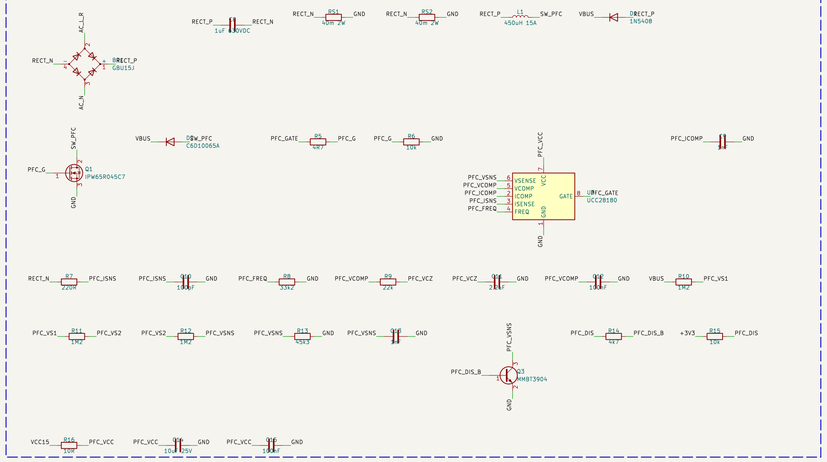
1 AC INPUT / INRUSH



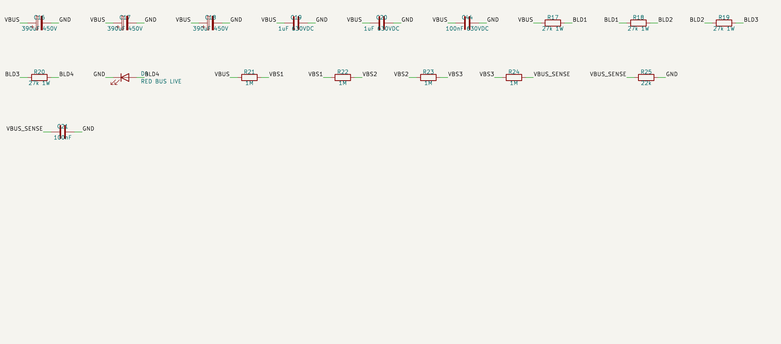
2 AUX SUPPLIES



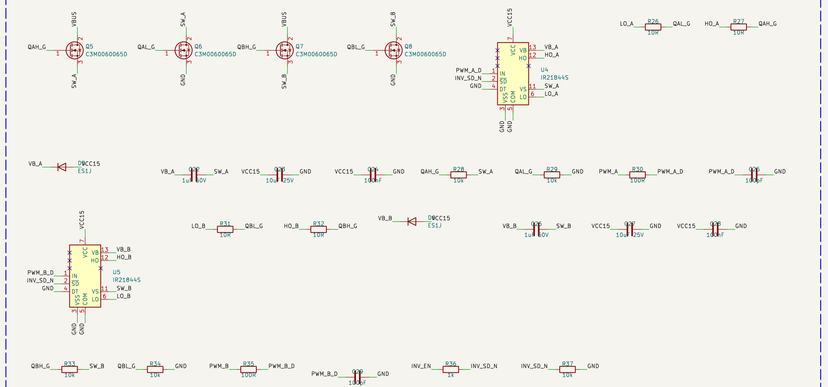
3 RECTIFIER + PCF



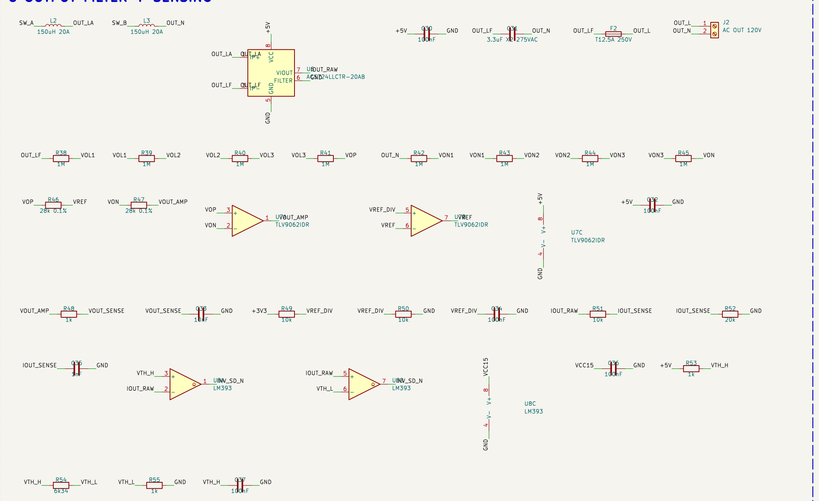
4 DC BUS 400V



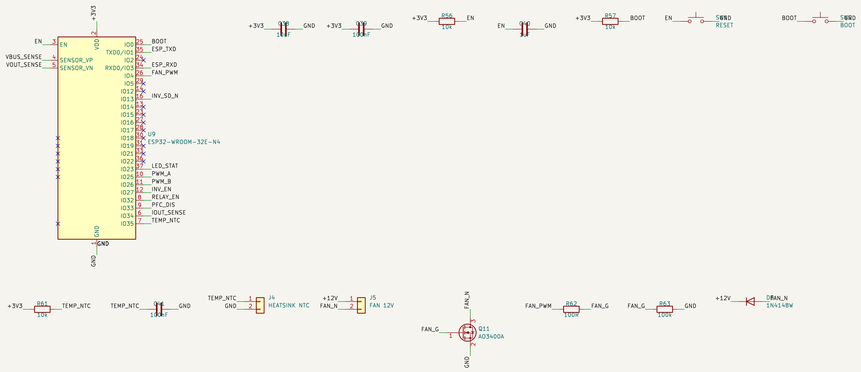
5 INVERTER BRIDGE



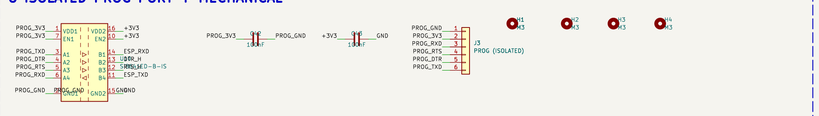
6 OUTPUT FILTER + SENSING



7 ESP32 CONTROLLER



8 ISOLATED PROG PORT + MECHANICAL



PWR_FLAG markers (ERC only)



230 VAC 50 Hz –> 120 VAC 50 Hz, 1200 W continuous. Non-isolated AC-DC-AC.
Bus GND is mains-referenced (HOT). Never connect a scope ground or USB to this board without isolation.
J2 output is NOT isolated from mains and has no neutral: do not bond either output pin to PE.
UCC28180 compensation (C9, R9, C11, C12) and R8 (fsw) are starting values: verify with TI design calculator.
Power devices (BR1, Q1, D2, Q5–Q8) mount on one heatsink with insulating pads; heatsink to PE.

DANGER: whole board is at mains potential, incl. ESP32 GND. Only J3 (prog) is isolated.
Generated from hardware/gen/design.py – edit there and regenerate

Sheet: /	
File: inv1200.kicad_sch	
Title: 230V –> 120V 1200W AC-DC-AC converter	
Size: A1	Date: 2026-10-12
KiCad E.D.A. kicad-cli 7.0.11-dfsg-1build4	Rev: A
	Id: 1/1